Coverage, Entry, and Engineering Components of Electricity Recovery in Japan's Municipal Waste-Incineration Fleet, FY2005–FY2024

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Abstract

Facility counts can misstate electricity-recovery coverage. We link 23,593 Japanese municipal-incinerator records for fiscal year (FY) 2005–FY2024 into 1,690 audited stable administrative lineages and 1,767 asset episodes, then examine count-versus-volume coverage, first reporting of installed electrical capacity, and components of gross megawatt-hours per tonne (MWh/t). In FY2024, 41.1% of records reported installed capacity, positive-output facilities handled 80.1% of throughput, and installed-capacity facilities held 70.5% of processing design capacity. Entry was sparse: 55 descriptive events, 35 broad exact-year events, and 33 after positive prior-year operation. Firth bias-reduced models show a strong positive association with processing scale: conditional odds ratios comparing 300 with 100 tonnes per day are 6.13 and 6.25. Lineage-bootstrap joint age tests do not support a general independent association in the broad, prior-operation, or identity-certain frames ($p = 0.380$, 0.186, and 0.357); a 24-event same-episode sensitivity is continuity-sensitive ($p = 0.051$ with 499 lineage-bootstrap replications). Among 6,511 engineering-valid generator-years across 493 lineages, reported start-year cohorts differ sharply in generator sizing. In a heating-controlled diagnostic, adding design intensity makes former age, processing-scale, and waste-utilization associations with gross MWh/t small and non-significant. Gross MWh/t is therefore an accounting outcome, not an independent operating-performance measure. An exploratory pathway comparison places continuity-lineage entrants below rebuild/replacement-like entrants in descriptive first-complete-year component ranks. The evidence shows count-volume divergence, scale-associated entry, and a reported start-year cohort hierarchy in generator sizing, but not physical project histories or causal effects.

Keywords: municipal solid waste; incineration; waste-to-energy; Japan; generator sizing; capacity factor; administrative record linkage

1 Introduction

Japan's municipal waste system relies extensively on incineration for volume reduction and hygienic treatment. Electricity recovery can use part of the heat released during combustion, but it does not by itself make incineration preferable to prevention, reuse, or recycling. Its value depends on plant design, waste properties, energy use, and the wider waste and electricity systems (Astrup et al., 2009, 2015; Brunner & Rechberger, 2015; European Commission, 2017). Japan-specific work likewise extends beyond a binary label of whether a plant generates electricity (Uno, 2015; Tabata & Tsai, 2016; Yamada et al., 2023).

The Ministry's FY2024 national summary reports 415 electricity-generating facilities among 991 incineration facilities, or 41.9% (Ministry of the Environment Japan, 2026). That published context is distinct from this paper's analytical measure: 417 of 1,014 retained FY2024 facility records report positive installed electrical capacity, or 41.1%. The numerator definition and record denominator differ, so the official ratio is not substituted into the analytical calculations.

The most visible national statistic is often a facility count. Yet a small intermittent facility and a large continuous facility each contribute one unit, despite handling different waste volumes. A count describes how widely equipment appears across records, not how much waste passes through generating facilities or where processing capacity is located. Those are distinct coverage questions.

A second problem arises in facility histories. First reporting of installed capacity is uncommon and can occur within an administratively continuous facility, with a reported asset reset, or in a forward-dated record. These cases do not have one verified physical meaning. The workbooks also lack a reliable identifier over the full period: codes are absent in FY2010–FY2012 and fully change between FY2019 and FY2020. Treating them as persistent keys would break histories and create false events.

A third problem concerns gross electricity divided by throughput. This MWh/t ratio combines generator size relative to processing capacity, annual use of installed electrical capacity, and waste loading. Treating it as an independent operating measure can make a design-cohort difference look operational.

This paper addresses the three problems in one national facility-level study. It reconstructs stable administrative lineages before creating lags, separates facility participation from waste-volume and design-capacity coverage, uses a bias-reduced event-history model for sparse first-entry events, and decomposes gross generation intensity into observable engineering components. The aim is descriptive and diagnostic. The study does not estimate the effect of a specific equipment project, infer a physical closure from administrative absence, or calculate net electricity export, useful heat, lifecycle emissions, or recoverable technical potential.

1.1 Research questions

The three research questions (RQs) are:

> **RQ1: Count versus waste-volume coverage.** How did the share of facility records reporting installed electrical-generation capacity evolve from FY2005 to FY2024, and how does that count-based measure compare with the share of recorded waste throughput handled by positive-output facilities and the share of waste-processing design capacity located at installed-capacity facilities?

> **RQ2: First reported installed-capacity entry.** Among stable administrative lineages first observed without installed electrical-generation capacity, which prior-year age and waste-processing-scale profiles are associated with first reporting positive capacity, and does the evidence change after the risk set is restricted to lineages with positive prior-year waste throughput?

> **RQ3: Engineering components and exploratory pathway outcomes.** Among operating generators, how do generator design intensity, electrical capacity factor, and waste-processing utilization combine to produce gross MWh/t; how are those components associated with reported design vintage and processing scale; and how do

first-complete-year component ranks differ descriptively between continuity-lineage and rebuild/replacement-like entries?

The sequence matters. RQ1 establishes the denominator problem at fleet level. RQ2 then asks which observed non-generating lineages enter the installed-capacity state. RQ3 begins only after positive throughput and gross output are observed, and asks what produces differences within the generating segment. Figure 1 shows the annual count, throughput, and design-capacity coverage measures that anchor this sequence.

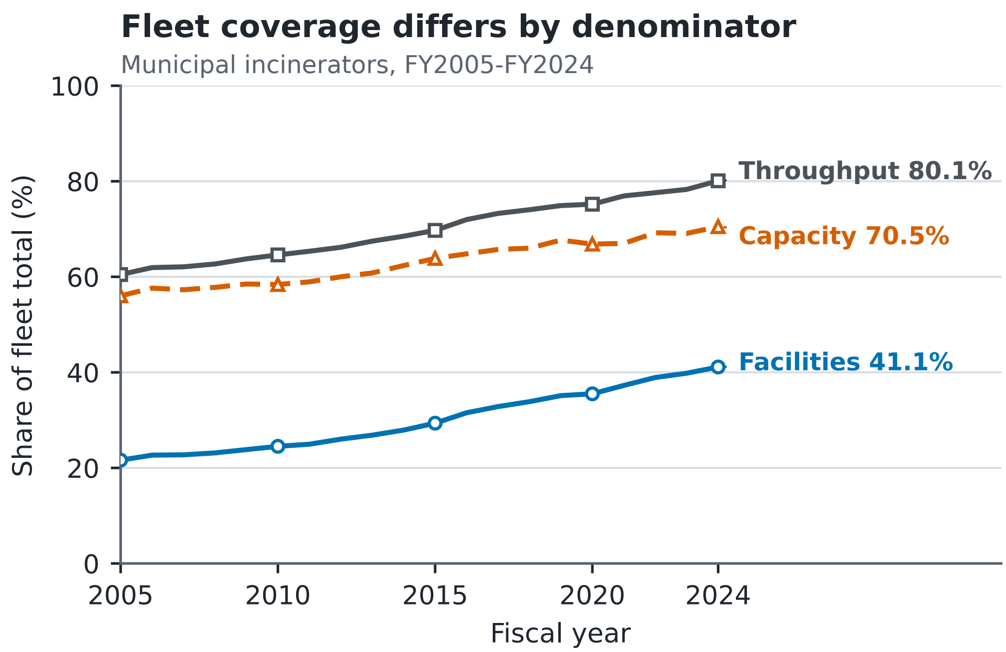

Figure 1: Facility participation, positive-output throughput coverage, and installed-capacity design-capacity coverage in Japan's municipal-incineration records, FY2005–FY2024. The denominators are intentionally different and should not be read as interchangeable shares.

2 Analytical Foundation and Comparator Adaptation

2.1 From one fleet average to three estimands

System studies consider material flows, lifecycle effects, energy substitution, and waste hierarchy (Astrup et al., 2009, 2015; Brunner & Rechberger, 2015; Münster & Meibom, 2010). Facility studies benchmark operating plants (Chen et al., 2012; Yeh, 2020), while Japan studies examine policy, power systems, heat use, and technology (Sasao, 2018; Shino, 2019; Tabata & Tsai, 2016; Uno, 2015). These perspectives do not share one denominator or estimand.

The present design separates three estimands. Fleet coverage is an annual ratio whose denominator is either facilities, tonnes, or waste-processing design capacity. Entry is a discrete-time conditional probability among lineages observed at risk. Generator components are conditional associations among positive-throughput, positive-output generator-years that pass stated engineering checks. A coefficient from the generator frame cannot explain why a non-generator enters, and a facility participation rate cannot describe the share of waste processed with electricity output.

This separation permits a low count share and high throughput share to be true simultaneously. It also permits newer cohorts to report higher gross MWh/t because of larger generators even when older generators do not have lower annual capacity factors. Section 3 states that identity explicitly.

2.2 What is adapted from high-profile and close comparators

High-profile studies supply research logic, not a ready-made model. Cui et al. (2026) foreground facility hierarchy, Liu et al. (2025) emphasize effectiveness over expansion, and Han et al. (2025) place recovery beside other sustainability dimensions. This paper asks where hierarchy appears in Japan without estimating an optimization frontier, city energy-carbon system, or pollutant outcomes.

Sasao (2018) supports repeated Japanese facility observations and explicit output questions. Shino (2019) makes generation relative to waste input an important observable. This paper retains that observable but treats it as an accounting intensity requiring decomposition.

Chen et al. (2012) and Yeh (2020) motivate separating activities within operating plants. This study adds an entry risk set and uses component regressions because the available fields do not support a comparable network or revenue frontier across all years.

Table 1: Transparent adaptation of comparator research logic

Comparator	Research logic used here	Adaptation in this paper	Outside this paper's estimand
Cui et al. (2026)	Examine hierarchy rather than only a fleet mean	Separate generator design intensity from annual capacity factor and waste loading	Plant optimization or a transferable frontier
Liu et al. (2025)	Distinguish effective system coverage from equipment expansion	Compare facility participation with throughput and design-capacity coverage	Urban energy-carbon effectiveness or welfare effects
Han et al. (2025)	Keep resource recovery distinct from other sustainability outcomes	State gross electricity boundaries and avoid a total-sustainability score	Pollutant-control, health, or lifecycle assessment
Sasao (2018)	Use repeated Japanese facility data for policy-relevant output questions	Construct a first-entry event history and explicit risk sets	Replication of the original policy specification
Shino (2019)	Treat electricity per waste input as an informative observable	Decompose gross MWh/t into sizing, capacity factor, and waste loading	Treating the ratio as an independent engineering score
Chen et al. (2012); Yeh (2020)	Recognize multiple plant activities and persistent heterogeneity	Model observable components and within-year ranks	Data-envelopment or revenue-frontier estimation

The adaptation changes the unit and estimands to fit Japanese administrative data. Its distinctive elements are audited lineages, count-volume coverage, sparse first entry, and an exact engineering identity; its conclusions are narrower than studies with process, cost, emissions, or city-system data.

3 Data and Methods

3.1 Source workbooks and recoverable provenance

The source is the Ministry of the Environment Japan General Waste Treatment Survey facility workbook series, also distributed through the Japanese official statistics system (e-Stat, n.d.; Ministry of the Environment Japan, 2026). The study covers 20 fiscal years, FY2005–FY2024. For each year, the checked-in parser reads the first workbook sheet and searches the first six spreadsheet rows for recognized Japanese header terms. The resulting standardized fields include facility and municipality information, reported start year, waste-processing design capacity, annual throughput, furnace and facility attributes, installed electrical capacity, and gross electricity generated.

The provenance audit records all 20 workbooks, their filenames, byte sizes, 256-bit Secure Hash Algorithm (SHA-256) hashes, parsed sheet names, detected header rows, and final field mappings. The files total 15,158,836 bytes and have 20 distinct hashes. Seventeen standardized fields are detected in the older workbooks; 19 are detected from FY2018 onward. Sold-electricity and sales-revenue fields are not detected for FY2005–FY2017, which is one reason the outcome is gross generation rather than net export or revenue. The original download timestamp was not recorded. Volatile checkout modification times are deliberately not persisted; the manifest instead records each workbook's last Git commit timestamp as repository history, not retrieval time. Configured source URLs were not revalidated during the provenance build. The hashes establish which local files produced the results, not the publisher's revision history or a complete acquisition chain.

Parsing yields 23,599 raw rows. Six rows are exact duplicates of another source record and are collapsed before identity resolution, leaving 23,593 unique facility-year records. Derived age is fiscal year minus reported start year. Thirty-six source ages are missing and 355 are negative. Negative values are converted to missing for age-dependent analysis rather than set to zero. Waste-processing utilization is annual throughput divided by 365 times design capacity. Installed generation is indicated by reported electrical capacity greater than zero.

3.2 Stable administrative lineages and asset episodes

Official facility codes cannot support the longitudinal design by themselves. All 3,716 rows in FY2010–FY2012 lack a code. Between FY2019 and FY2020, both years contain codes, yet the sets have zero overlap. The latter break is a complete recode, not evidence that the entire fleet was replaced. Across that transition, the identity resolver restores 1,064 adjacent-year lineage links, equivalent to 97.3% of FY2019 records. Across the longer FY2009–FY2013 bridge, 882 official codes overlap while 1,135 stable lineages are linked.

The resolver therefore constructs a *stable administrative lineage*: a sequence of records judged to describe the same continuing administrative facility or site history. The implementation field is `stable_site_id`, but the term does not assert that ownership, buildings, furnaces, or other physical assets remain unchanged. The algorithm is deterministic and proceeds as follows.

1. Text and identifiers are normalized conservatively, including Unicode form, spacing, punctuation, and numeric-code formatting. A complete standardized row fingerprint identifies exact duplicate records, which are collapsed before matching.

2. Records are compared only within prefecture. Candidate evidence combines normalized facil-

ity name, municipality, reported start year, waste-processing capacity, furnace count, facility type, and the annual official code. Code agreement is supporting evidence, but contradictory name and configuration evidence can veto a code match.

3. Adjacent fiscal years are resolved before gaps, preventing an older history from winning a tie over an otherwise equivalent immediately prior record. Short gaps of up to four years are considered afterward. Within each prefecture-year problem, one-to-one global assignment prevents one prior record from being assigned to multiple current records.

4. Unmatched records seed new deterministic lineage IDs from canonical record fingerprints rather than row order. Within a lineage, a new asset episode is created when reported start year changes materially in either direction, a mature record resets to a near-zero age, or a major name and configuration discontinuity appears.

The result contains 1,690 stable administrative lineages and 1,767 asset episodes, with no duplicate lineage-year and no history longer than the 20-year study window. The match audit classifies 15,324 records as code plus exact-name links, 274 as code with other supporting evidence, 6,204 as exact-name links without code support, 101 as fuzzy-name links without code support, and 1,690 as new lineage seeds.

Candidate links below the minimum score are excluded before assignment, as are ambiguous links lacking an exact-name or official-code signal. This removes 3,092 sub-threshold and 15,308 weak ambiguous candidate edges. Sixteen accepted links, spanning 14 lineages, still have a low current-record or prior-record margin but retain strong evidence; every one is exposed in the identity audit. The entry and component analyses therefore include whole-lineage identity-certain sensitivities that exclude all 14 affected lineages rather than selectively removing individual years.

Known difficult cases are embedded as executable checks. Three pairs that must link and three pairs that must remain separate are tested directly, including examples from the FY2019–FY2020 recode and earlier code reuse. The resolver is also rerun after random row permutation and after insertion of an unrelated synthetic record in six difficult or duplicate-bearing prefectures. The lineage/episode mapping must remain unchanged. These golden-link, separation, permutation, and insertion tests reduce identifiable implementation risks; they do not make probabilistic record linkage certain. Any result depending on a small number of lineages remains vulnerable to unresolved name or configuration ambiguity. The uncertainty flag makes that vulnerability testable rather than implicit.

3.3 Analytical frames and estimands

The three RQs use related but non-identical samples. Table 2 prevents their denominators from being blended.

The entry sample excludes 467 lineages already reporting positive installed capacity in their first observed year because their entry time is left-censored. A remaining lineage contributes annual risk rows until its first positive capacity observation or its last observed non-generating year. The first observed risk row is dropped from modeling because a lagged predictor is not available. The complete-covariate frame further drops 120 rows with missing lagged age or processing capacity, and 22 non-adjacent rows are excluded from the exact-year model; none of those 22 rows contains an event.

The 55 descriptive events, the 35 complete-covariate events, and the pathway categories answer

Table 2: Analytical frames after identity and data-quality checks

Frame	Facility-year rows	Stable lineages	Events	Estimand
Full administrative panel	23,593	1,690	N/A	Annual facility, throughput, and design-capacity coverage
Descriptive non-generator risk set	16,519	1,223	55	Observed first installed-capacity entries during follow-up
Broad exact-year complete-covariate entry model	15,154	1,137	35	Conditional annual administrative-lineage entry odds using prior-year covariates
Exact-year model after prior operation	13,072	1,019	33	Nested sensitivity requiring positive prior-year throughput
Same-asset-episode continuity sensitivity	15,095	1,135	24	Entry odds after excluding transitions that cross an inferred asset-episode boundary
Identity-certain-lineage sensitivity	15,107	1,130	35	Broad model after excluding every lineage containing an accepted uncertain identity link
Positive-throughput, positive-output generators	6,660	504	N/A	Descriptive operating-generator frame
Engineering-valid component model	6,511	493	N/A	Conditional generator-component associations

different bookkeeping questions. Coincidentally, the pathway audit also identifies 35 continuity-lineage events. That count should not be conflated with the 35 events in the exact-year regression sample. The former is a pathway classification among observed entries; the latter is a covariate completeness restriction.

3.4 RQ1: coverage definitions and fleet identity

For fiscal year t, let N_t be all retained facility records, I_{it}^K indicate positive installed electrical capacity, I_{it}^G indicate positive gross generation with positive throughput, W_{it} be annual throughput, and C_{it} be waste-processing design capacity. The three headline coverage measures are

$$P_t^{\text{facility}} = \frac{\sum_i I_{it}^K}{N_t}, \tag{1}$$

$$P_t^{\text{throughput}} = \frac{\sum_i W_{it} I_{it}^G}{\sum_i W_{it}}, \tag{2}$$

and

$$P_t^{\text{design}} = \frac{\sum_i C_{it} I_{it}^K}{\sum_i C_{it}}. \tag{3}$$

The first uses facility records, the second tonnes, and the third tonnes per day of design capacity. Positive output is used in the throughput numerator because installed capacity does not guarantee positive annual generation. For the engineering-valid subset, fleet gross intensity has the exact decomposition

$$\frac{\sum_i G_{it}^{\text{valid}}}{\sum_i W_{it}} = \left(\frac{\sum_i W_{it}^{\text{valid}}}{\sum_i W_{it}}\right)\left(\frac{\sum_i G_{it}^{\text{valid}}}{\sum_i W_{it}^{\text{valid}}}\right). \tag{4}$$

The first term is valid generator-throughput coverage and the second is throughput-weighted conditional gross intensity. This identity distinguishes how much waste is covered from how much gross electricity is reported per tonne within the covered segment.

3.5 RQ2: sparse first-entry model

The event is the first observed year with positive installed electrical capacity after a lineage has been observed without it. It is an administrative state transition. It does not uniquely identify first physical operation, equipment installation, or a particular construction history.

Discrete-time event-history analysis represents each at-risk lineage-year as a binary outcome (Allison, 1982; Beck et al., 1998). For lineage i in year t, the main model is

$$\begin{aligned}\operatorname{logit}\{\Pr(Y_{it} = 1 \mid Y_{i,t-1} = 0)\} = \alpha + \beta_A' A_{i,t-1} + \beta_C \log\left(1 + \frac{C_{i,t-1}}{100}\right) \\ + \gamma' E_t + \delta' D_{it}.\end{aligned} \tag{5}$$

Here, A contains age bands 10–19, 20–29, and 30 or more years relative to 0–9 years; C is prior-year processing design capacity in t/day; E contains four calendar eras; and D contains flexible elapsed-risk-duration bands of 1–4, 5–9, 10–14, and at least 15 years. The capacity transform is scaled so its contrast remains interpretable without assuming a linear effect per t/day. The specification contains 11 coefficients but only 35 complete-covariate events, or approximately 3.2 events per coefficient. It is therefore intentionally limited to age, processing scale, calendar era, and elapsed risk duration. Adding technology or geographic terms would further spend sparse event information without creating causal control.

Ordinary maximum-likelihood logit is vulnerable to small-sample bias and separation in a sparse event design. The primary estimator therefore uses Firth's bias reduction (Firth, 1993) and its finite-estimate solution to separation (Heinze & Schemper, 2002), maximizing the penalized log-likelihood

$$\ell_F(\theta) = \ell(\theta) + \frac{1}{2}\log|\mathcal{I}(\theta)|, \tag{6}$$

where ℓ is the binomial log-likelihood and $\mathcal{I}$ is the expected information matrix. The Jeffreys-prior penalty reduces first-order bias and keeps finite estimates under separation. The coefficient table labels the fitted-model standard errors, confidence intervals, and term-level p-values as model-based. To represent repeated observations from the same lineage, the primary uncertainty calculation uses 499 deterministic cluster-bootstrap replications that resample complete lineages with replacement. Percentile intervals use the resulting coefficient distributions. Joint age tests apply a Wald statistic to the covariance of the three age coefficients across the same 499 bootstrap replications. Every requested replication must converge and return all focal coefficients; otherwise the analysis fails rather than silently dropping a bootstrap draw.

Four frames are fitted. The broad exact-year frame includes every complete at-risk row whose lag belongs to the same stable administrative lineage; it can cross an inferred asset-episode boundary because its estimand is first administrative-lineage entry. The prior-operation frame is a nested sensitivity that additionally requires positive prior-year throughput. It is not an independent comparison group, so differences between it and the broad frame are not treated as an equality or equivalence test. A same-asset-episode continuity sensitivity excludes the 59 exact-year rows that cross an inferred episode boundary, including 11 events. An identity-certain sensitivity excludes every lineage containing any of the 16 accepted uncertain links. These two additional frames show whether inference depends on the continuity and linkage rules.

For an intuitive scale contrast, the odds ratio comparing 300 with 100 t/day is

$$\begin{aligned} OR_{300:100} &= \exp[\beta_C \{\log(1 + 300/100) - \log(1 + 100/100)\}] \\ &= \exp(\beta_C \log 2). \end{aligned} \tag{7}$$

Conventional logit and complementary-log-log fits are retained only as link sensitivity checks.

3.6 RQ3: engineering decomposition and component models

Let G_{it} be annual gross generation in MWh, W_{it} annual throughput in tonnes, K_{it} installed electrical capacity in kW, and C_{it} waste-processing design capacity in t/day. Define

$$Y_{it} = \frac{G_{it}}{W_{it}} \qquad \text{(gross generation intensity, MWh/t),} \tag{8}$$

$$D_{it} = \frac{K_{it}}{C_{it}} \qquad \text{(generator design intensity, kW per t/day),} \tag{9}$$

$$F_{it} = \frac{G_{it}}{8.76K_{it}} \qquad \text{(annual electrical capacity factor),} \tag{10}$$

$$U_{it} = \frac{W_{it}}{365C_{it}} \qquad \text{(waste-processing utilization).} \tag{11}$$

The factor 8.76 converts one kW operating for 8,760 hours into annual MWh. These definitions imply the exact facility-year identity

$$Y_{it} = \frac{8.76}{365}\frac{D_{it}F_{it}}{U_{it}} = 0.024\frac{D_{it}F_{it}}{U_{it}}. \tag{12}$$

Gross MWh/t is thus a ratio produced jointly by installed sizing, annual use of that electrical capacity, and annual waste loading. The identity does not say that any one component is exogenous. For example, throughput, capacity factor, maintenance, and waste composition may be jointly determined during a year.

The primary sample requires positive installed capacity, throughput, and gross output. Values are excluded rather than clipped if gross intensity falls outside 0.01–0.80 MWh/t, electrical capacity factor outside 0.02–1.20, waste-processing utilization outside 0.02–1.20, or generator design intensity outside 0.1–100 kW per t/day. A non-missing non-negative reported age and all model fields are also required. Of 6,660 positive-throughput, positive-output rows, 149 fail at least one check, leaving 6,511 rows across 493 lineages. Heating value between 3 and 25 megajoules per kilogram (MJ/kg) is audited as a plausibility field but is not an inclusion condition for the primary component models; 102 otherwise valid rows lack it.

Two pooled component models are estimated by ordinary least squares with standard errors clustered by stable lineage (Wooldridge, 2010):

$$\log D_{it} = \alpha_D + V_{it}'\beta_D + \beta_{DC}\log C_{it} + T_{it}'\eta_D + \lambda_t + \varepsilon_{Dit}, \tag{13}$$

$$\begin{aligned}\log F_{it} = \alpha_F &+ V_{it}'\beta_F + \beta_{FC}\log C_{it} + \beta_U U_{it} \\ &+ T_{it}'\eta_F + \lambda_t + \varepsilon_{Fit}.\end{aligned} \tag{14}$$

V_{it} contains reported start-year cohorts before 1990, 1990–1999, and 2000–2009 relative to 2010 or later. Reported start year is an administrative design-vintage marker, not a verified boiler or generator installation date. T includes furnace count and coarse furnace and facility-type groups; λ_t are fiscal-year indicators. A direct gross output model replaces $\log C$ with $\log W$ and $\log K$, retaining cohort, technology, furnace-count, and year terms.

A specification diagnostic compares a legacy-style regression of $\log Y$ on reported age, processing scale, waste utilization, heating value, technology, and year with the same model after adding $\log D$. Both specifications use the 5,806 engineering-valid rows with reported heating value in the 3–25 MJ/kg plausibility range. This is not a causal mediation design. It asks whether coefficients previously attributed to age, scale, or utilization are stable after the omitted installed-sizing component is represented.

Robustness checks split the period into FY2005–FY2014 and FY2015–FY2024, give each lineage equal total weight, and apply conservative and broad predefined engineering bounds. Asset-episode fixed effects and exact-adjacent first differences are used only for operating components that vary meaningfully within an episode. Design intensity is predominantly a between-asset attribute, so within-episode change is not presented as its primary estimand.

Finally, event pathways are classified from observed administrative continuity. A continuity-lineage entry remains in the same lineage and asset episode across adjacent years without a reported reset. A rebuild/replacement-like entry has an asset-episode, reported-start-year, or mature-to-new age reset. A forward-dated/placeholder entry has a future start year or new-build placeholder name. These labels are descriptive evidence rules. They do not verify the physical project mechanism. First-complete-year outcomes are compared as within-fiscal-year percentile ranks among engineering-valid generators, which reduces confounding by fleet-wide time trends but does not remove selection into each pathway.

4 Results

4.1 RQ1: facility counts understate waste-volume coverage

The official 415/991 ratio (41.9%) describes the Ministry's published count of electricity-generating facilities. The results below use the separate analytical definition of positive installed capacity among retained facility records.

Installed-capacity participation rises steadily from 21.6% of facility records in FY2005 to 41.1% in FY2024. Positive-output facilities already handle a much larger share of waste: throughput coverage rises from 60.5% to 80.1% over the same period. The design-capacity share at installed-capacity facilities moves from 56.0% to 70.5%. Figure 1 therefore shows long-run increases with year-to-year variation under all three denominators, alongside a persistent count-volume gap.

The FY2024 cross-section makes the distinction concrete. The administrative panel contains 1,014 facility records, of which 417 report positive installed electrical capacity, giving the 41.1% participation rate. There are 410 positive-throughput, positive-output facilities. They process 24.70 million of the 30.84 million recorded tonnes, or 80.1%. In contrast, 469 operating non-generators process 6.14 million tonnes, or 19.9%. Another 126 records report neither positive throughput nor generation, and nine installed-capacity records report no positive output. Because output and installed-capacity flags are not identical, these segment counts should not be forced into a single binary partition.

After the predefined output and capacity-factor checks, valid generator rows cover 79.7% of FY2024 throughput. Their throughput-weighted conditional gross intensity is 0.425 MWh/t. Multiplying 0.797 by 0.425 gives 0.338 MWh per total fleet tonne, exactly matching valid gross generation divided by all recorded throughput. This arithmetic is more informative than the facility count alone: most recorded waste is already processed at facilities reporting positive generation, even though installed equipment appears in fewer than half of facility records.

The result does not imply that the remaining 19.9% of operating throughput can or should all be redirected or equipped. It also does not measure electricity used internally, net export, heat recovery, marginal emissions, or project cost. Its contribution is denominator discipline: 41.1%, 80.1%, and 70.5% answer different questions, and no one of them is a sufficient measure of the remaining opportunity.

4.2 RQ2: entry is rare and strongly associated with processing scale

The descriptive risk set contains 55 first reported installed-capacity entries. The pathway audit classifies 35 as continuity-lineage entries, 11 as rebuild/replacement-like, and nine as forward-dated or placeholder entries. In the bridge to reported output, 47 of the 55 show positive gross generation in the event year and 51 do so by the following observed fiscal year. Installed capacity and positive annual output are therefore closely related but not identical states.

Only 35 events remain after requiring an exact one-year lag and complete prior age and capacity; 33 remain after requiring positive prior-year throughput. This sparse count is central to interpretation. The Firth model addresses finite-estimate and first-order bias problems, but it cannot create information that the panel does not contain.

Observed rates already show strong scale ordering. Entry occurs in 1 of 3,854 risk rows in the smallest processing-capacity quartile (0.026%), 2 of 4,175 in the second quartile (0.048%), 9 of 3,702 in the third (0.243%), and 23 of 3,423 in the largest (0.672%). Because risk duration, calendar period, and age differ across these groups, the adjusted model is needed before interpreting the gradient.

In the broad exact-year Firth model, the coefficient on $\log(1 + C/100)$ is 2.6158. In the prior-operation frame it is 2.6444. Both 499-lineage-bootstrap intervals are wholly positive: 1.9707 to 3.4872 and 1.9087 to 3.7149, respectively. The implied 300-versus-100 t/day odds ratios are 6.13 and 6.25. Because the prior-operation frame is nested, the two coefficients are parallel sensitivity estimates rather than a valid between-group contrast. Conventional logit and complementary-log-log capacity coefficients, 2.6258 and 2.6091, are close to the broad Firth estimate.

Age requires more caution. Relative to age 0–9, broad-frame coefficients are -1.254 for age 10–19, -1.127 for age 20–29, and -1.309 for age 30 or more. Their broad-frame cluster-bootstrap intervals each include zero. Prior-operation point estimates are similarly negative, and its three percentile intervals also include zero. The coefficients are correlated, so the designated summary is the joint test based on lineage-bootstrap covariance. The age terms are not jointly significant in the broad frame ($\chi^2 = 3.08$, 3 df, $p = 0.3800$), the prior-operation frame ($\chi^2 = 4.81$, 3 df, $p = 0.1863$), or the identity-certain frame ($\chi^2 = 3.24$, 3 df, $p = 0.3566$).

The same-episode continuity result is more fragile. With only 24 events, its joint test based on 499 lineage-bootstrap replications gives $\chi^2 = 7.78$ (3 df, $p = 0.0508$), whereas the corresponding fitted-model covariance gives $p = 0.0099$. The discrepancy, rather than either threshold classification, is the relevant finding: age inference is sensitive both to the continuity definition and to within-lineage dependence. It is not evidence of a stable general age effect. Figure 2 therefore presents only the broad and prior-operation estimates for readability; the same-episode and identity-certain coefficient sets remain in the supplement and generated evidence.

The supported RQ2 conclusion is narrow but clear. First reported capacity entry is rare and strongly concentrated among larger waste-processing facilities. The administrative data do not establish a general independent age pattern: the broad and identity-certain results are null, while the near-threshold same-episode result depends on a stricter continuity rule and sparse events. Age should not be used as if the model had identified an equipment-project response.

4.3 RQ3: the gross-intensity hierarchy is principally a sizing hierarchy

The engineering-valid generator frame has 6,511 observations from 493 stable lineages. Median gross intensity is 0.327 MWh/t, median generator design intensity is 14.0 kW per t/day, median electrical capacity factor is 0.607, and median waste-processing utilization is 0.609. These summaries describe generator-years that pass stated bounds, not the full fleet.

Reported start-year cohorts differ much more in generator sizing than in annual capacity factor. Median design intensity rises from 5.33 kW per t/day before 1990 to 10.83 in the 1990s, 15.83 in the 2000s, and 20.59 in 2010 or later. Median gross intensity rises in parallel from 0.145 to 0.283, 0.348, and 0.475 MWh/t. Median electrical capacity factors are 0.619, 0.625, 0.561, and 0.664; they do not show a comparable monotonic vintage gradient. Figure 3 separates these two components visually.

The adjusted component models reinforce the descriptive pattern. Relative to the 2010-or-later cohort, log generator design intensity coefficients are -1.565 before 1990, -0.883 for 1990–1999, and

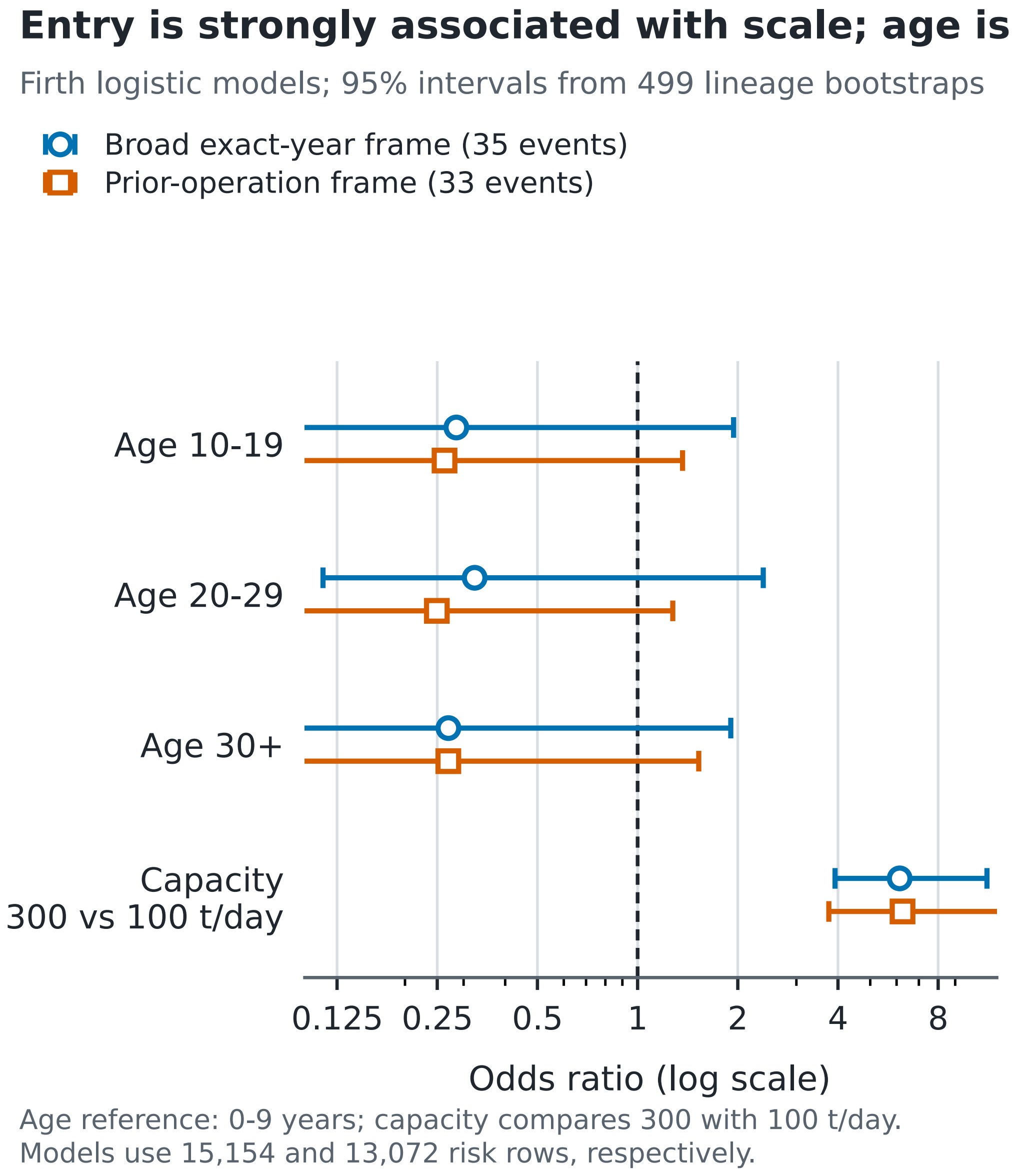

Figure 2: Firth bias-reduced estimates for first reported installed-capacity entry in the broad exact-year and prior-operation frames. Capacity is shown as the 300-versus-100 t/day contrast; intervals come from 499 stable-lineage bootstrap replications.

-0.267 for 2000–2009, all with $p < 0.001$. At the same processing capacity and observed controls, these imply substantially smaller installed generators in older reported cohorts. The elasticity of generator design intensity with respect to processing design capacity is 0.532; its 95% confidence interval (CI) is 0.447 to 0.617. The model explains 54.9% of variation in log design intensity.

The electrical-capacity-factor model tells a different story. Conditional on processing scale, waste utilization, technology, furnace count, and fiscal year, the pre-1990 and 1990s coefficients are positive, 0.302 and 0.198, relative to 2010 or later. The 2000s coefficient is 0.015 ($p = 0.606$). Waste utilization is positively associated with log electrical capacity factor (coefficient 1.695, 95% CI 1.448 to 1.942), while processing scale is negatively associated (-0.116, 95% CI -0.162 to -0.070). This

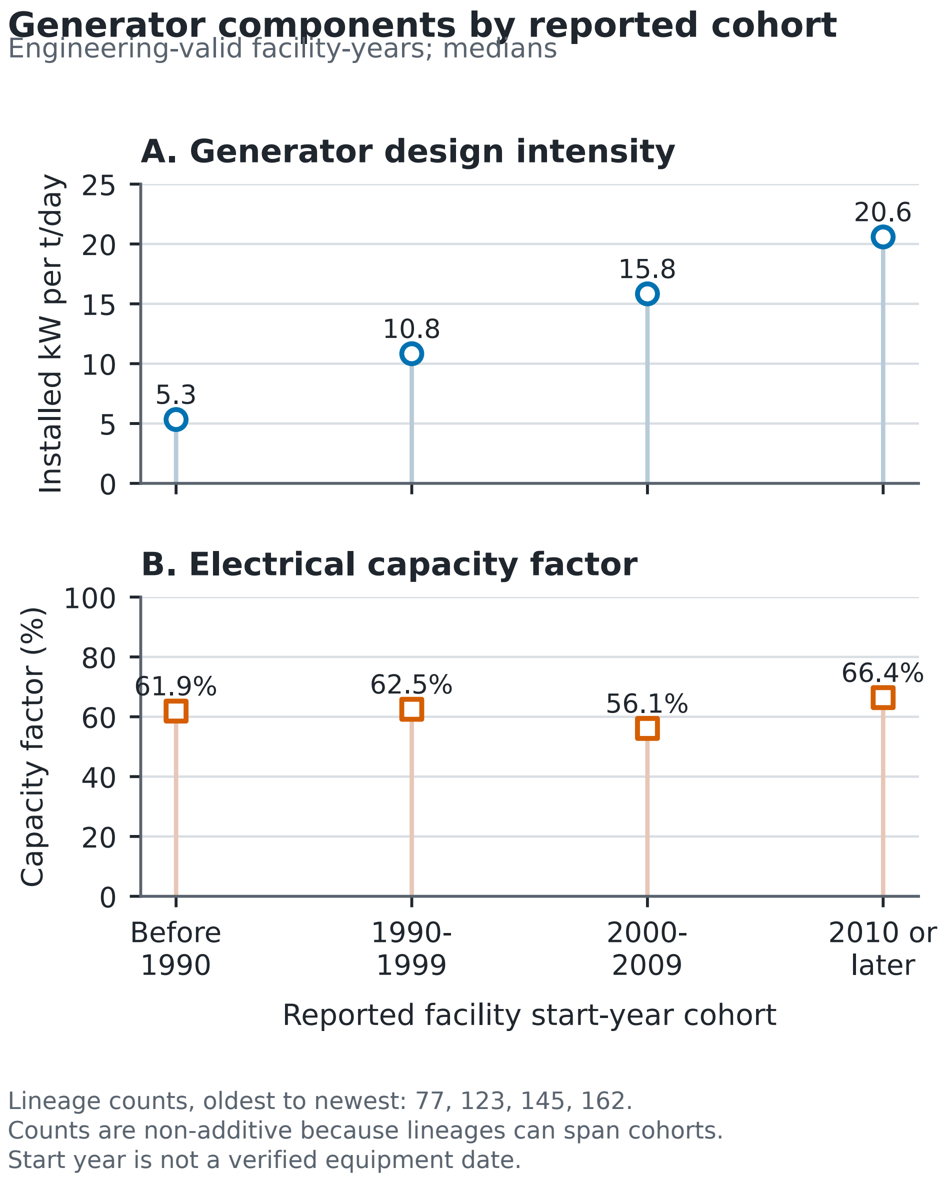

Figure 3: Median generator design intensity and electrical capacity factor by reported facility start-year cohort among engineering-valid generator-years. Reported start year is a design-vintage marker, not a verified generator installation date.

model explains 33.9% of log capacity-factor variation. These coefficients describe annual use of installed kW. They do not show that utilization has an independent positive association with gross MWh/t after generator sizing is represented.

The direct gross-output model is consistent with the component structure. The elasticity of annual gross MWh with respect to throughput is 0.638 (95% CI 0.536 to 0.740), and the elasticity with respect to installed electrical capacity is 0.576 (95% CI 0.502 to 0.650), conditional on cohort, observed technology, furnace count, and year. The model R^2 is 0.914. It confirms that both waste loading and installed kW are central to annual output, but it is not a production function with exogenous inputs.

The most important diagnostic compares gross-intensity specifications on the 5,806-row plausible-heating-value subset. In the legacy-style model without generator design intensity, the reported-age coefficient is -0.0349, the processing-capacity coefficient is 0.1001, and the waste-utilization coefficient is 0.6699; all have $p < 0.001$. After adding log generator design intensity, the corresponding coefficients become -0.0020 ($p = 0.2977$), -0.0092 ($p = 0.1991$), and -0.0995 ($p = 0.2038$). The sizing coefficient is 0.7532 ($p < 0.001$), and model R^2 rises from 0.4737 to 0.8131. This is not a causal decomposition. It shows that the former age, scale, and utilization interpretation is sensitive to whether installed generator sizing is represented.

Adjacent-year rank persistence is highest for generator design intensity: 0.995 across 5,963 pairs from 470 lineages. Gross-intensity rank persistence is 0.961 and capacity-factor persistence is 0.873. The corresponding within-to-total variance ratios are 0.016 for log design intensity, 0.089 for log gross intensity, and 0.426 for log capacity factor. Sizing is therefore mostly a between-asset design attribute, while annual capacity factor contains much more within-asset movement.

The principal component conclusions remain stable across the two ten-year windows, lineage-equal weighting, and conservative or broad engineering bounds. For example, the processing-scale coefficient in the design-intensity model is 0.474 in FY2005–FY2014 and 0.577 in FY2015–FY2024; it is 0.520 under conservative bounds and 0.536 under broad bounds. Excluding every identity-uncertain lineage leaves 6,450 rows across 487 lineages and gives a nearly unchanged coefficient of 0.533. Within-asset-episode models also retain a positive association between utilization and electrical capacity factor. Those checks strengthen the component description but do not resolve simultaneous changes in throughput, maintenance, installed capacity, and output.

4.4 Exploratory pathway comparison

The first-complete-year comparison asks where entrants sit in the same-year generator distribution, not what entry causes. Across all exact-year entrants with available engineering-valid outcomes at event time plus one, 44 lineages have mean gross-intensity rank 51.6%, generator-design rank 48.1%, and capacity-factor rank 56.3%. The pooled average is therefore near the middle of the contemporaneous generator distribution.

Pathways are heterogeneous. At event time plus one, 27 continuity-lineage entrants average 0.260 MWh/t and rank at 40.2% for gross intensity, 36.8% for generator design intensity, and 53.8% for electrical capacity factor. Eleven rebuild/replacement-like entrants average 0.442 MWh/t and rank at 72.5%, 66.1%, and 65.5%, respectively. Figure 4 displays these component ranks. Six forward-dated/placeholder observations are omitted from the plotted contrast because that category is too sparse and its administrative timing is difficult to interpret.

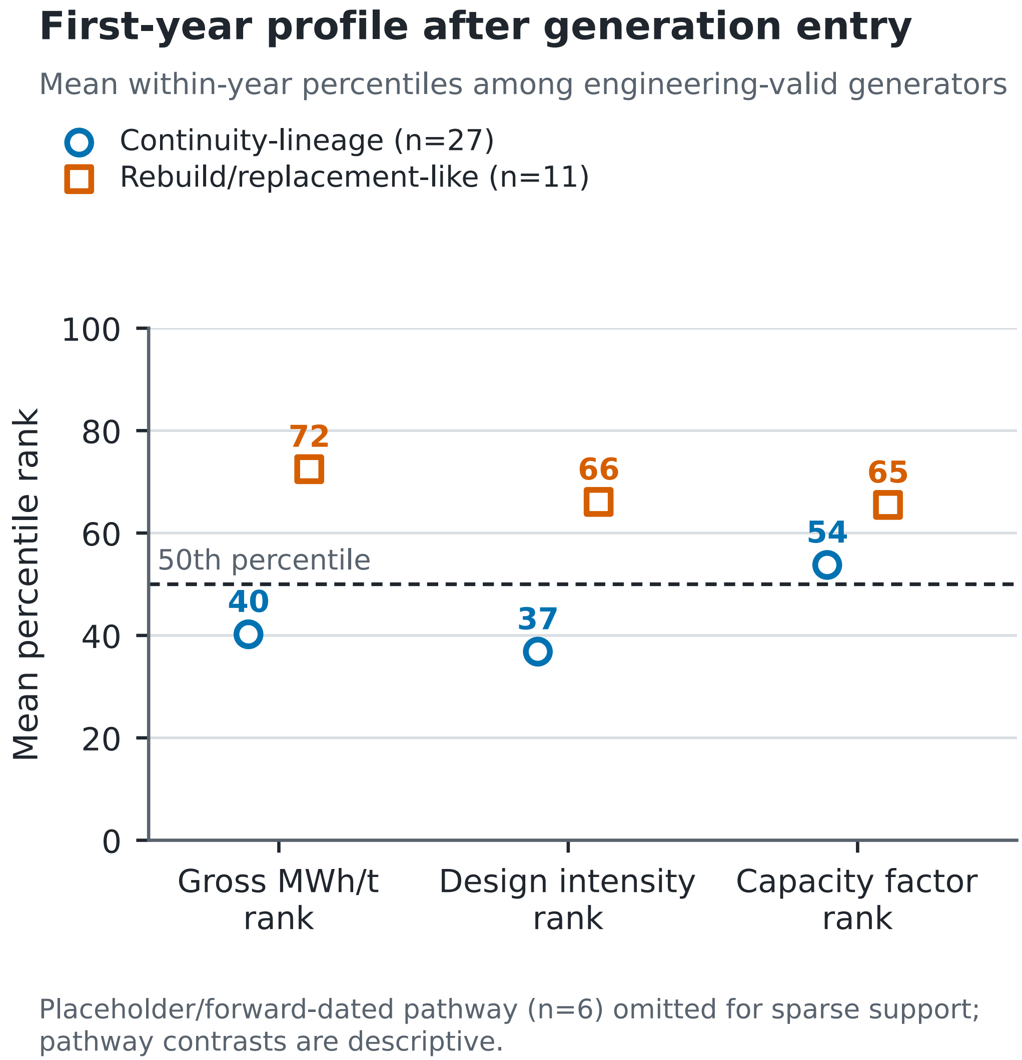

Figure 4: First-complete-year mean within-year percentile ranks for continuity-lineage and rebuild/replacement-like entrants. The contrast is descriptive; the sparse forward-dated/placeholder pathway is omitted.

The larger pathway difference aligns more closely with generator sizing than with capacity factor. That pattern is consistent with the broader component results, but it is not an estimated effect of pathway. Pathway assignment is based on reported resets, follow-up is selective, and there is no counterfactual match between otherwise equivalent projects.

5 Discussion

5.1 What the paper changes in the fleet narrative

The 41.1% analytical participation rate describes equipment distribution, not waste coverage. Positive-output facilities handle 80.1% of throughput, and the installed segment holds 70.5% of processing design capacity. Japan therefore combines incomplete diffusion across records with substantial volume coverage; non-generating records are not equal-sized unused opportunities.

Entry is strongly associated with processing scale in both frames, but the model does not show that enlarging a facility would produce entry. Scale can proxy for unobserved municipal, technical, and financial conditions. Age is weaker evidence: individual estimates are negative, broad and identity-certain joint tests are non-significant, and the same-episode result lies at the conventional threshold with fewer events. This sensitivity is evidence against a simple age-only narrative, not evidence for one preferred continuity definition.

Gross MWh/t also changes meaning after decomposition. Its observed cohort hierarchy aligns primarily with generator sizing; older reported cohorts do not have uniformly lower annual capacity factors. Once sizing is included, the other legacy coefficients no longer carry a distinct gross-intensity interpretation. This is the value of adapting comparator work on hierarchy and multi-activity production while respecting the narrower Japanese fields.

5.2 Evidence-bound implications

Monitoring should report facility, throughput, and design-capacity coverage together. Screening can treat processing scale as a marker for further feasibility work, but not use an age-only rule. Existing generators should be compared conditional on sizing and design vintage before gross MWh/t is interpreted as an operating gap.

These are measurement and diagnostic implications, not a ranked list of projects. The study does not determine whether a municipality should build, replace, coordinate, maintain, or retire a facility. Such decisions require capital costs, waste-supply agreements, grid and internal-use conditions, maintenance history, emissions controls, heat demand, and alternatives higher in the waste hierarchy. The paper's role is to prevent an aggregate statistic or misspecified ratio from deciding those questions implicitly.

5.3 Limitations and next evidence needed

The 20 workbooks are hash-identified and parser mappings are documented, but their original retrieval timestamps and publisher-side revision history are unavailable. Future acquisition should archive both.

Stable administrative lineages remain inferred. The resolver addresses exact duplicates, code breaks, row-order dependence, and known difficult links, but no physical-site registry confirms ownership, construction, or closure histories. Administrative absence is therefore not modeled as a physical outcome.

Entry is rare and partially left-censored. Firth estimation and lineage bootstrapping cannot overcome only 35 broad-frame, 33 prior-operation, and 24 same-episode events. Although all 499 requested bootstrap replications converge, resampling cannot create missing project information. The parsimonious models omit financing, prices, contracts, catchments, maintenance, and detailed technology. Scale may proxy for these, so odds ratios are associations rather than effects of changing t/day.

Generator fields also have strict boundaries. Gross generation is not net export; on-site use and useful heat are incomplete; reported capacity may differ from availability; and MWh/t is not the

European Union R1 energy-efficiency indicator, a lifecycle measure, or an economic measure (Grosso et al., 2010). Heating-value coverage is insufficient for a common thermal measure. Bounds remove 149 rows but may exclude unusual valid cases or retain plausible-looking errors.

Reported start year is not an equipment date and can bundle original design, later equipment, reporting, waste, and municipality context. Annual throughput, capacity factor, maintenance, and output are jointly determined. Controls, fixed effects, and adjacent differences do not create exogenous variation; the equations remain accounting identities and conditional descriptions.

Finally, the pathway contrast is small and selected, and reported resets do not verify physical replacement. Linking procurement, construction, generator, net-export, heat-use, outage, and waste-composition records would enable the project-specific questions this panel cannot answer.

6 Conclusion

Japan's municipal-incineration fleet looks different when the denominator and engineering structure are made explicit. In FY2024, installed capacity appears in 41.1% of facility records, while positive-output facilities handle 80.1% of throughput and installed-capacity facilities represent 70.5% of processing design capacity. First reported capacity entry is sparse and strongly associated with processing scale, with 300-versus-100 t/day odds ratios near six in both modeled frames. Broad and identity-certain age tests are non-significant, while a same-episode sensitivity uses only 24 events and differs sharply between bootstrap and fitted-model covariance; age inference is therefore continuity-sensitive rather than generally established. Among operating generators, the observed gross-MWh/t cohort hierarchy aligns primarily with generator sizing and design vintage; it does not support an independent operating or age interpretation once sizing is included. An exploratory comparison places continuity-lineage entrants below rebuild/replacement-like entrants in first-complete-year component ranks, but that small, selected contrast remains descriptive.

The paper's contribution is therefore not a claim that every non-generator is an equal opportunity or that newer facilities simply perform better. It is a defensible measurement architecture: reconstruct lineages before transitions, separate counts from waste volume, use sparse-event methods for entry, and decompose gross output before interpreting hierarchy. That architecture gives a professor or later reviewer a clear foundation for deciding which next question requires engineering, capital-history, or causal evidence.

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Pann Phetra: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval

This study uses publicly released administrative facility data and does not involve human participants, animal subjects, or private personal data.

Data Availability

The facility-level source data are derived from the Ministry of the Environment Japan General Waste Treatment Survey, which is publicly released by the Ministry of the Environment. Subject to source-file redistribution conditions, the author will deposit the analysis code, machine-readable result manifests, derived tables, figure-generation scripts, and manuscript figures in a versioned repository associated with the submitted paper. Raw source workbooks can be obtained from the Ministry and e-Stat portals.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this manuscript, the author used OpenAI Codex and Anthropic Claude for language revision, manuscript organization, and assistance with code development and review. The author executed the analyses, inspected the source data and generated outputs, independently checked the reported results, reviewed and edited all AI-assisted material, and takes full responsibility for the content. These tools were not used as authors and did not replace author judgment or accountability.

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